

# Introduction

# Objectives of Software Testing

- Find defects / prevent defects
- Good Quality
- Meet product requirements
- Fit for use

# Manual Testing

- Analyze Requirements
- Test Planning
- Test Case design
- Test case execution
- Test report

# Seven Principles of Software Testing

- Early testing
- Testing shows presence of defects
- Exhaustive testing is not possible
- Defect clustering
- Pesticide Paradox
- Testing is context dependent
- Absence of error fallacy

# Software Development Lifecycle

- Requirement Gathering and Analysis
- Design
- Development
- Testing
- Deploy to production
- Maintenance

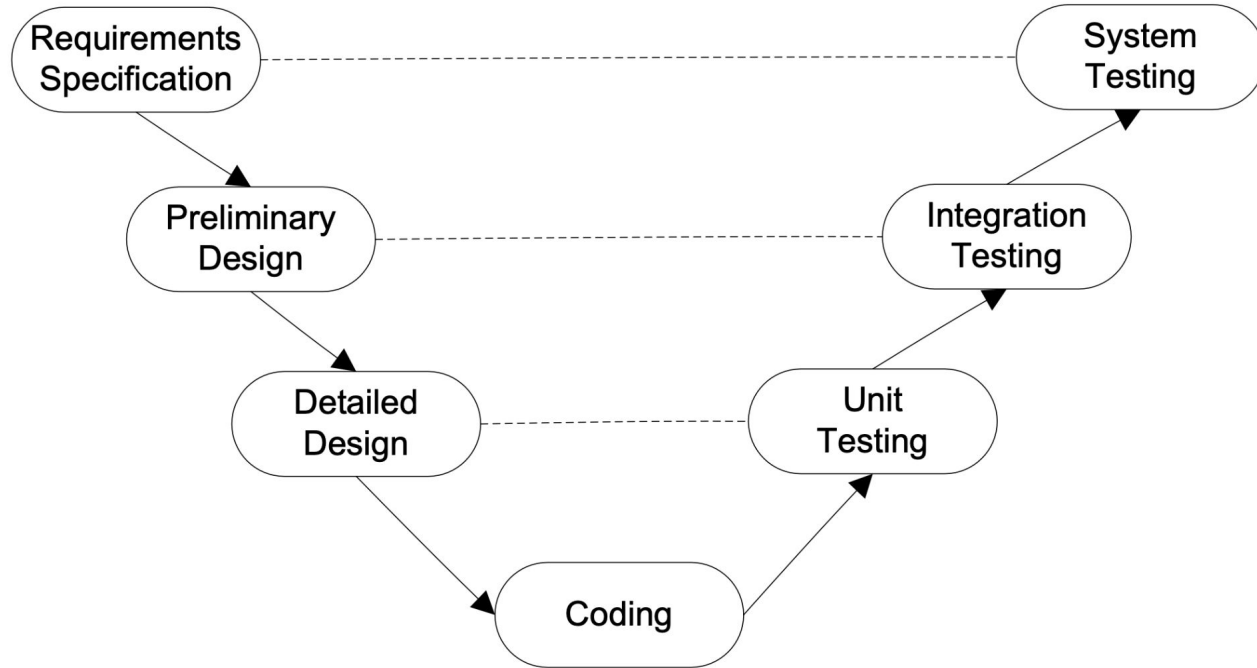
# Software Testing Life Cycle

- Requirement analysis
- Test planning
- Test case design & development
- Test environment setup
- Test execution
- Test reporting
- Test closure

# Waterfall model

- Requirement Gathering and Analysis
  - Global Design
    - Detailed design
      - coding
        - Testing
          - Deploy to production
            - Maintenance

# What is V model?





# Iterative development model

- Phase 1 (iteration duration 1 or 2 months)
  - Requirement
    - Design
      - Development
        - Testing
          - Deploy
- Phase 2 (iteration duration 1 or 2 months)
  - Requirement
    - Design
      - Development
        - Testing
          - Deploy
- Phase 3 (iteration duration 1 or 2 months)
- ...
- Phase N

Phase 1 focuses on feature 1

Phase 2 focuses on 2nd feature

Phase N focuses on last feature

# Levels of Testing

- Unit Testing / Component Testing
- Integration Testing
- System Testing
  - System testing (whole standalone system test)
  - System integration testing (external integration test)
- User acceptance Testing

## Functionality of an ecommerce website

- Registration requirement
- Add item to the cart
- Checkout
- Cancel the order

# Types of Software Testing

- Functional (testing the functionality of a particular software)
  - Unit test, Integration test, System test, User acceptance test, regression test, sanity test, smoke test, usability test
- Nonfunctional (test the performance of the software such as response time, latency, throughput)
  - Performance testing, load testing, stress testing, volume testing, scalability testing, recovery testing, compatibility testing, security testing

# What is Functional Testing?

- Gmail service
- User requirements / market research
- Functionality/features?
  - Create account
  - Compose email
    - Edit email, send email
- Test case?
  - Click create account button and provide details and press create account button
  - Should pass all the steps

# What is Black Box Testing?

- Don't have knowledge of internal mechanism
- Match input and output
- Provide input and verify the output
- Gmail test:
  - Launch [gmail.com](https://www.gmail.com) in the browser; output should be the landing page of the website
  - Click on create account; output should be create account page
  - After providing the account information; output should redirect to the actual gmail account page

# White Box testing

- What happens in the actual code
- What are the classes, what are the functions, methods
- Calculator application example
  - Check the parameter data type, return data type
  - Check add(), division(), multiplication() functions and pass the
  - Statement coverage, branch coverage, condition coverage, loop coverage, path coverage
- Can happen in both unit and integration tests

# Black box vs white box

| <b>Black Box</b>                                       | <b>White Box</b>                                         |
|--------------------------------------------------------|----------------------------------------------------------|
| Done by software testers                               | Done by developers                                       |
| Do not need to know the internal code                  | Need to know the internal code and implementation        |
| Verifying the functionality and requirement            | Verifying the structure of the technical design document |
| Do not need to know the programming language           | Need to know the programming language                    |
| Regressing testing, system testing, acceptance testing | Statement coverage, condition coverage, loop coverage    |
|                                                        |                                                          |



# What is non-functional testing?

- Tests the responsiveness and stability of the system
- How long does it take to load or redirect the page after clicking the button?
- Processing time: how long does it take to process the payment for the customer?
- Concurrent user volume: how does the system handle lots of users?

## Comment non-functional Testing

- **Load Testing:** up to 5K users and response time should be less than 5 sec
- **Stress Testing:** when the system breaks down if we keep increasing the load
- **Soak Testing:** running the load test for an extended period of time such as for a week
- **Volume Testing:** testing the load on database; how the systems handles lots of data during the peak hours
- **Spike Testing:** how the system handles spike of users

# Functional vs Non-functional Testing

| Functional testing                                | Non-functional test                              |
|---------------------------------------------------|--------------------------------------------------|
| Checks whether the functionalities work or not    | Checks the performance of the functionality      |
| Focuses on user requirements                      | Focuses on user expectations                     |
| Can be done manually; also can be automated       | Difficult to done manually, need to be automated |
| Unit testing, integration testing, system testing | Load testing, spike testing, volume testing      |
|                                                   |                                                  |

# What is Agile testing?

- Duration with 2 to 4 weeks (scrum framework)
- Communication between business and product owners
- Collaborating testing approach; whole team is responsible for the success
- collaborating with the development team; communicating the test scenarios
- Developers, testers, and business analysts are involved in the team meetings
- Quality is everyone's responsibility

# Principles of Agile Testing

- Whole team is responsible for the success and tests; not only the QA team
- Continuous testing; done in sprints
- Continuous feedback on the domes of the user stories
- Test driven development
- Less documentation; only the necessary information

# Methods of Agile testing

- Behavior Driven Development (BDD)
- Acceptance Test Driven Development (ATDD)
- Exploratory Testing
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